

MINSK NATL, Belarus

WMO: 261148

Lat: 53.882N

Lon: 28.031E

Elev: 670

StdP: 14.34

Time Zone: 3.00 (E03)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.0	-0.1	-9.8	3.3	-5.8	-4.0	4.6	1.2	19.9	20.5	18.1	20.9	3.8	310	0.573

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	85.6	67.5	82.1	66.3	78.7	65.0	70.2	80.1	68.3	77.3	66.8	75.4	7.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	100.3	74.0	64.8	94.1	71.8	64.0	91.5	70.9	34.6	80.2	33.0	77.3	31.7	74.5	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.8	16.4	14.3	DB	-13.3	90.0	7.1	3.1	-18.4	92.2	-22.6	94.0	-26.6	95.8	-31.8	98.1	(4)
(5)			WB	-13.6	73.0	7.0	2.0	-18.7	74.4	-22.8	75.6	-26.7	76.7	-31.9	78.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.0	22.1	23.9	32.1	44.7	54.6	61.0	65.5	63.4	54.0	43.3	35.0	27.6		
	DBStd	17.10	11.44	10.93	8.80	7.47	7.79	6.21	5.48	6.05	6.59	7.56	7.69	9.34		
	HDD50	3802	866	732	556	191	42	2	0	0	34	233	450	696		
	HDD65	7831	1331	1152	1020	608	332	152	62	106	334	673	900	1161		
	CDD50	1626	0	0	1	33	185	332	480	416	153	25	1	0		
	CDD65	179	0	0	0	0	10	32	77	56	4	0	0	0		
	CDH74	1949	0	0	0	9	150	358	788	592	52	0	0	0		
	CDH80	490	0	0	0	1	22	66	213	184	4	0	0	0		
	WSAvg	6.4	6.8	6.9	7.2	6.6	6.4	6.1	5.6	5.3	5.5	6.1	6.8	7.3		
	PrecAvg	24.8	1.6	1.5	1.5	1.5	2.4	2.9	3.6	2.4	2.0	2.1	1.8	1.6		
(15) Precipitation	PrecMax	34.0	3.1	3.0	2.5	3.1	4.6	6.2	5.7	6.8	4.8	5.6	3.6	2.8		
	PrecMin	20.1	0.3	0.7	0.3	0.1	0.6	0.8	0.3	0.2	0.4	0.1	0.3	0.6		
	PrecStd	3.8	0.7	0.6	0.6	0.7	1.1	1.3	1.5	1.5	1.1	1.3	0.8	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.2	42.8	58.8	74.8	82.7	86.0	89.2	91.5	78.9	67.8	53.9	46.8		
		MCWB	42.7	41.1	47.9	57.4	64.8	67.5	68.7	66.5	64.0	58.2	50.8	45.6		
	2%	DB	39.3	39.4	52.2	68.4	78.6	82.1	85.7	84.6	75.0	61.2	50.1	43.1		
		MCWB	38.6	38.0	43.5	54.1	62.7	65.6	68.0	66.9	62.0	55.3	48.5	42.5		
	5%	DB	37.3	37.5	48.1	64.0	73.8	78.6	82.2	80.5	69.9	57.4	47.8	41.0		
		MCWB	36.4	36.4	41.7	51.5	60.0	64.4	67.1	65.7	60.1	53.6	46.6	40.4		
	10%	DB	34.2	35.6	43.2	59.2	70.0	75.0	78.6	76.6	66.1	53.9	44.7	37.8		
		MCWB	33.8	34.6	39.6	48.5	57.8	62.9	66.0	64.6	58.0	51.0	43.8	37.2		
	0.4%	WB	42.8	41.0	49.5	59.2	67.8	71.1	72.9	72.1	67.0	59.9	52.0	46.1		
		MCDB	43.1	41.9	56.9	71.0	78.1	80.5	82.8	83.6	75.1	65.3	53.5	46.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.2	38.1	45.3	55.3	64.4	68.3	70.8	69.5	63.2	56.8	48.7	42.8		
		MCDB	38.6	38.7	50.5	65.0	74.4	77.3	80.6	79.4	71.1	60.4	49.8	42.9		
	5%	WB	36.0	36.2	42.5	52.9	62.0	66.3	68.7	67.5	60.8	54.1	46.0	40.1		
		MCDB	36.5	36.9	45.9	61.0	71.6	75.1	78.0	76.2	67.7	57.4	46.9	40.6		
	10%	WB	33.9	34.4	39.8	50.3	59.6	64.0	67.1	65.5	58.9	51.8	43.6	37.2		
		MCDB	34.3	34.9	42.9	56.6	67.4	71.7	75.5	73.9	64.8	54.4	44.3	37.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.7	14.1	20.3	21.3	20.6	19.8	20.2	18.6	13.0	7.5	7.2		
		MCDBR	6.5	8.6	20.3	27.9	26.7	25.9	25.4	26.1	24.7	18.0	10.6	7.6		
		MCWBR	6.4	7.2	13.0	15.0	13.1	12.2	10.8	11.2	13.5	12.4	9.0	7.2		
		MCDBR	6.6	7.5	17.3	24.2	23.8	22.6	21.9	22.1	22.0	16.4	9.4	7.2		
	5% WB	MCWBR	6.6	6.8	12.0	13.9	12.7	12.0	10.8	11.0	13.1	12.0	8.6	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.323	0.355	0.409	0.400	0.385	0.414	0.417	0.378	0.352	0.331	0.291		
	taud		2.165	2.119	2.159	2.189	2.288	2.357	2.287	2.276	2.346	2.398	2.281	2.202		
	Ebn at Noon		206	237	257	259	269	274	263	254	250	230	191	180		
	Edh at Noon		21	30	36	41	39	37	39	37	30	23	19	16		
(44) All-Sky Solar Radiation	RadAvg		214	474	895	1291	1648	1783	1678	1429	961	485	174	125		
	RadStd		43	90	139	155	163	143	164	132	85	96	26	35		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+1.35	N/A	N/A	N/A	N/A	N/A	N/A	-418	+225	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	+1.46	N/A	N/A	N/A	N/A

Nomenclature: See separate page